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Article

Keywords:

Posted Date: February 25th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-8902418/v1

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Additional Declarations: There is **NO** Competing Interest.

Theory-Informed Generative Agents for Human Mobility Modeling

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ABSTRACT

Human mobility follows robust population-level regularities, yet individual behavior remains highly heterogeneous and context-dependent. The advances of generative agents (large language model (LLM)–driven, persona-conditioned computational agents) offer a promising approach to modeling rich, individualized behavior, but they often lack theoretical grounding and do not readily scale to population-level simulation due to the computational cost of generating decisions for large numbers of agents over long horizons. To reconcile mechanistic understanding with flexible, human-like decision modeling, we introduce a theory-informed mobility agent (TIMA) framework that integrates the physical constraints of mobility theory with the semantic reasoning of LLMs. Mechanistically, the LLM parameterizes the decision tendencies within the physical mobility model process to capture both the scaling laws of aggregate flows and the heterogeneity of individual preferences. Unlike prior LLM agents that are typically evaluated on small, bespoke samples, TIMA enables population-scale simulation while reconstructing key mobility patterns and diverse activity portfolios across multiple U.S. cities in a data-light manner. Beyond pattern reconstruction, we leverage TIMA to investigate mobility-mediated social phenomena. Specifically, we examine how individual choices shape experienced segregation and drive behavioral adaptations in response to external shocks. Using the COVID-19 pandemic as a case study, we demonstrate that the framework naturally captures shifts in segregation and reductions in activity as a function of evolving disease risk and policy constraints. Ultimately, this work illustrates a paradigm for integrating physical theory with generative AI, providing a scalable and flexible foundation for the next generation of policy-relevant behavioral simulation.

Introduction

Human mobility is a fundamental expression of human behavior, driven by the need to access resources, pursue opportunities, and maintain social life. Through countless individual decisions made in daily life, these movements collectively generate macroscopic mobility patterns that shape how societies function, from social development and segregation to economic activities and disease transmission. Modeling the interplay between these complex movements is therefore critical for designing resilient urban systems and effective public policy.

The universal patterns of human mobility have been extensively studied through theory-based models grounded in physics and statistical mechanics. A central insight is that, despite microscopic complexity, aggregate mobility obeys simple, transferable regularities. Early work formulated the interaction between two locations as a function of population size and distance, now conventionally referred to as the gravity law[1]. Building on this paradigm, advanced mechanistic approaches such as the intervening opportunities[2,3] and radiation models[4,5] were developed to capture more complex mobility patterns with concise mathematical forms. With the emergence of large-scale mobility datasets, subsequent studies uncovered universal statistical properties of human movement patterns[6], notably the heavy-tailed distributions of travel distance[7], Zipf-like distributions of visitation frequency[8], and log-normal distributions of trip duration[9]. Synthesizing these insights, mechanistic frameworks such as the Exploration and Preferential Return (EPR) family of models[10–12] encode those behaviors into a parsimonious stochastic process capable of reproducing empirical mobility patterns across locations. Collectively, these models provide an interpretable and transferable foundation for understanding human mobility at scale.

In parallel, a complementary perspective highlights the inherent heterogeneity of human mobility, in which movement decisions reflect how people interpret places, activities, and opportunities in their environment. These decisions vary across individuals and contexts, shaped by personal preferences, motivations, and constraints[13], and are inherently semantic and context-dependent. Capturing such variation requires encoding semantic information about individual intent and decision-making. Recent advances in large language models (LLMs) offer new opportunities to model these content-dependent aspects of mobility[14,15]. Trained on large-scale textual and multimodal data, LLMs enable reasoning about locations, activities, and personal interests in a human-like manner[16–18]. Pioneering studies have leveraged LLMs for generative agent-based mobility simulation[15,19–21], fine-tuning with spatial cognition[22–24], semantic reasoning over locations and activities[25,26], and personalized trajectory generation[27]. Together, these studies demonstrate the potential of LLM-based approaches to generate context-aware and adaptive movement behaviors at the individual level. However, the high computational and economic costs associated with LLM-driven agents currently limit most existing approaches to small-scale simulations, with agent populations rarely exceeding a few thousand, posing challenges for population-scale mobility modeling[28].

Taken together, the tension between universality and heterogeneity raises a natural question: how can the semantic reasoning captured by LLMs be integrated with physical mobility theory, so both perspectives can be captured simultaneously? Addressing this integration is essential not only for scalable, structurally grounded mobility simulation but also for enabling mechanism-driven analyses of mobility's social consequences and its reconfiguration under external shocks that are often unseen or poorly represented in training data (e.g., epidemics; policy interventions such as stay-at-home orders and closures). From a population structure perspective, models such as d-EPR [11] provide the essential spatiotemporal regularities, ensuring the population-level travel patterns follow universal physical laws. From an individual behavior perspective, LLMs' reasoning capabilities can translate unstructured context into explicit, agent-specific behavioral tendencies (e.g., purpose preferences, avoidance/attraction, compliance) that vary across individuals and settings[29,30], thereby addressing behavioral heterogeneity that traditional models cannot handle. Through combining the principled constraints of mechanism-driven models with the contextual adaptability of generative AI, we can model human mobility that conforms to physical laws at the macro level while remaining heterogeneous and expressive at the micro level.

This paper proposes Theory-Informed Mobility Agents (TIMA), a generative agent-based framework that integrates classical mobility theory with LLMs' reasoning capabilities to model human mobility (Fig. 1). In this framework, physical theory provides the structural backbone of the model. Specifically, we adopt the EPR (Explore and Preferential Return) model[10] to govern agents' exploration–return dynamics, while constraining travel

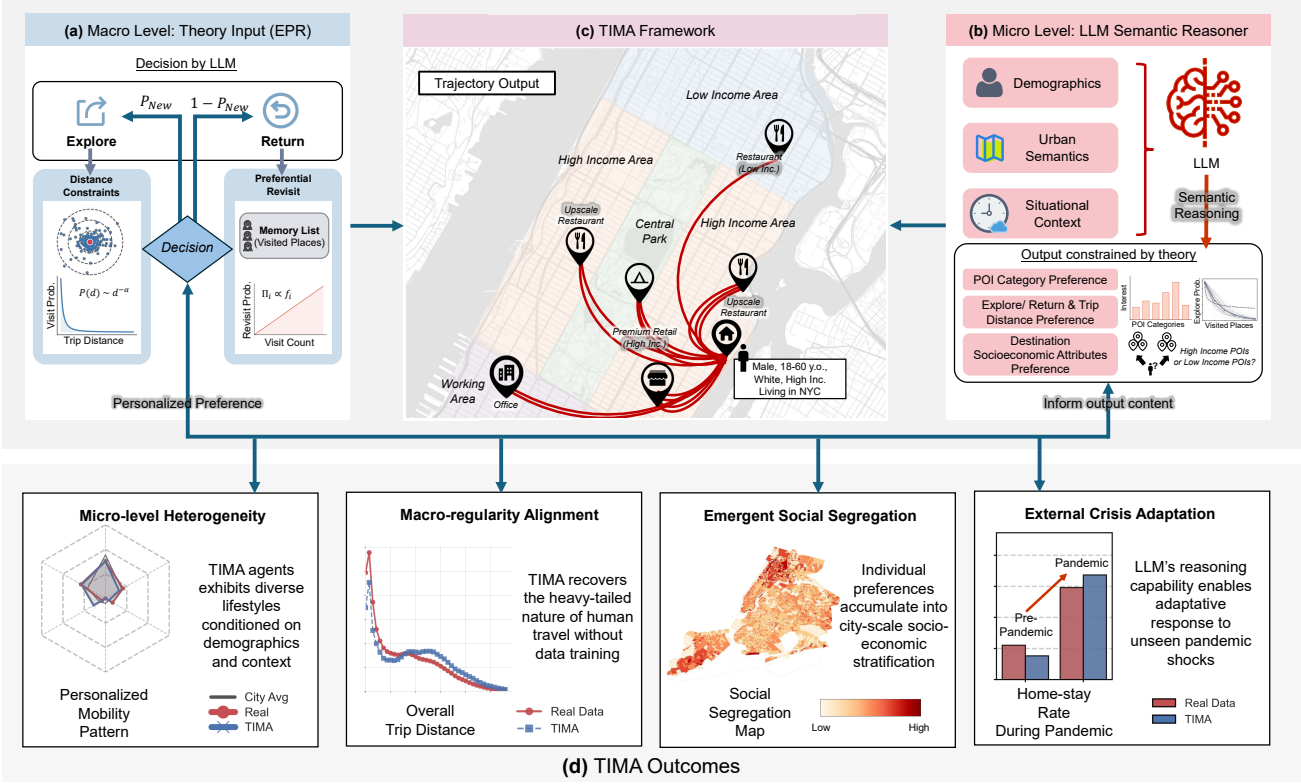


Figure 1. Overview of the Theory-Informed Mobility Agents (TIMA) framework. a, Macro-level structural backbone. Grounded in EPR theory, this layer establishes the physical constraints of human mobility, enforcing distance decay and exploration-return dynamics. **b,** Micro-level semantic reasoning. An LLM reasoner synthesizes an agent's demographic and socioeconomic attributes, urban semantics, and situational context to model adaptive mobility behavioral intent, including interests in different POI categories, tendency of exploration, and preference for travel distance and socioeconomic attributes of destinations. **c,** Trajectory output. The TIMA framework produces context-aware travel trajectories that distinguish visitation preference across socio-economic groups and POI categories (e.g., high-income individuals prefer to visit locations in high-income areas) and can identify the travel intent for different POI categories while being bounded by physical distance decay constraints. **d,** Multi-scale outcomes. TIMA successfully reproduces 1) individual lifestyle anisotropy by showing different demographic groups' diverse mobility patterns across POI categories, 2) macro-regularity alignment by reconstructing the heavy-tailed trip distance distribution, 3) social segregation phenomena by matching the stratification of urban visitation, and 4) counterfactual crisis adaptation by exhibiting emergent response to the external shock of the COVID-19 pandemic.

distances through distance-decay mechanisms. Built upon the theory, LLMs are used to incorporate individual heterogeneity by inferring context-dependent preferences for exploration, return, and destination choice based on agents' demographics and the local environments. By unifying structural constraints with semantic reasoning, TIMA achieves three advances: (1) reconstructs realistic mobility patterns across multiple U.S. cities without training on ground-truth mobility data, (2) provides a generative lens for studying mobility-mediated social phenomena such as experienced segregation and its reconfiguration under shocks, and (3) enables prospective counterfactual simulations of behavioral response during crises and rapid policy shifts. More broadly, this work illustrates a general paradigm for integrating physical theory and generative AI to support scalable and policy-relevant human behavior simulation[31].

Results

Fig. 1 illustrates the architecture of TIMA, which unifies population-scale physical structure with individual-level semantic reasoning. The framework operates across two distinct layers. In the structure layer, agents' behaviors are strictly governed by the classical EPR theory, with each step in an explore-return decision weighted by distance decay (Fig. 1a). In the cognitive layer (Fig. 1b), the LLM agents produce individualized preferences that shape exploration propensity and destination choice conditioned on demographics, local semantics, and situational context (see Methods Section "The Theory-Informed Mobility Agents framework" for example implementation). Mechanistically, at each step, the agent (1) instantiates agent-specific explore–return tendencies, (2) conditional on exploration, infers a feasible trip purpose (a POI category) under spatial constraints, and (3) selects a specific POI to visit by sampling a destination from LLM-inferred agent-CBG propensities combined with distance-penalized gravity friction (see Fig. 6 for prompt implementation). This yields a unified framework where personalized movement reasoning operates within the bounds of universal mobility laws.

We initialize TIMA with synthetic agents' personas that are representative of the real-world CBG population

Table 1. Performance comparison of mobility models across four U.S. metropolitan areas. The table reports the performance of TIMA against four baseline models (*d*-EPR, EPR, Social-EPR, Gravity) in New York City, Chicago, Orlando, and Norfolk-Virginia Beach. Evaluation Metrics: (a) Trip Distance (KL): The Kullback-Leibler divergence (KL) between simulated and observed trip-distance distributions. (b) OD Flow (CPC): The Commuter Public Component (CPC) overlap between simulated and observed Origin-Destination (OD) flows aggregated at the Census Tract level. (c) Visitation Density (MSE): Mean Squared Error (MSE) of the visitation density at the CBG level, computed on log-transformed values and normalized by the maximum log density to account for heavy tails and ensure comparability. (d) POI Proportion (KL): KL divergence between simulated and observed population-level proportions of visits assigned to each POI category at the CBG level. (e) Stratified OD Fidelity (CPC): Average CPC overlap of OD flows, computed separately for groups defined by the socioeconomic attributes of their origin home locations (e.g., income, education, and occupation; see Methods section "Evaluation protocol" for definitions). Arrows indicate whether lower (↓) or higher (↑) values are better.

City	Model	Trip Distance (KL) ↓	OD Flow (CPC) ↑	Visitation Density (MSE) ↓	POI Proportion (KL) ↓	Stratified OD Fidelity (CPC) ↑
New York City	TIMA (Ours)	0.054	0.289	0.006	0.181	0.231
	d-EPR	0.118	0.283	0.050	0.904	0.226
	EPR	0.408	0.253	0.007	1.135	0.206
	Social-EPR	0.393	0.180	0.007	1.384	0.148
	Gravity	0.432	0.246	0.006	0.763	0.202
Chicago	TIMA (Ours)	0.007	0.384	0.005	0.207	0.326
	d-EPR	0.009	0.402	0.026	0.570	0.343
	EPR	0.726	0.203	0.008	0.954	0.169
	Social-EPR	0.621	0.159	0.013	0.856	0.138
	Gravity	0.752	0.228	0.005	0.932	0.191
Orlando	TIMA (Ours)	0.042	0.550	0.008	0.827	0.462
	d-EPR	0.043	0.557	0.009	1.022	0.447
	EPR	0.570	0.437	0.014	0.963	0.385
	Social-EPR	0.564	0.410	0.014	0.776	0.363
	Gravity	0.501	0.378	0.015	0.974	0.303
Norfolk-Virginia Beach	TIMA (Ours)	0.049	0.522	0.009	0.555	0.390
	d-EPR	0.050	0.550	0.019	0.553	0.405
	EPR	0.521	0.424	0.017	0.769	0.316
	Social-EPR	0.518	0.380	0.012	0.797	0.299
	Gravity	0.451	0.381	0.009	0.566	0.284

across multiple socioeconomic characteristics, including gender, age, income, education, and occupation (see Methods section "Agent initialization" for details). Agents are embedded in a digital environment defined by each CBG's geographic boundaries, socioeconomic context, and observed POI distributions. The TIMA framework is scalable and generalizable across cities. In this study, we deploy it in four U.S. cities spanning diverse scales and morphologies (see detailed city-level information in Supplementary Information Section 1.2), including New York City, Chicago, Orlando, and Norfolk–Virginia Beach, to simulate general weekly mobility patterns. TIMA generates individual-level mobility trajectories that can be aggregated to the relevant spatial and temporal scales for direct comparison with data. We then compare the simulated outputs with empirical mobility data from Safegraph[32]. Crucially, TIMA does not access, train on, or fit to any real mobility data during simulation; the empirical data serve strictly as a ground-truth benchmark. To ensure the results are not dependent on a specific model architecture, we validate the framework using three distinct LLMs (Gemini-2.5-pro[33], GPT-4o[34], and Qwen-2.5-7b-Instruct[35]). The results presented in the main manuscript are modeled with Gemini-2.5-pro, while comparative results for other models are provided in the Supplementary Information Section 3.8.

Reconstruction of human mobility across cities

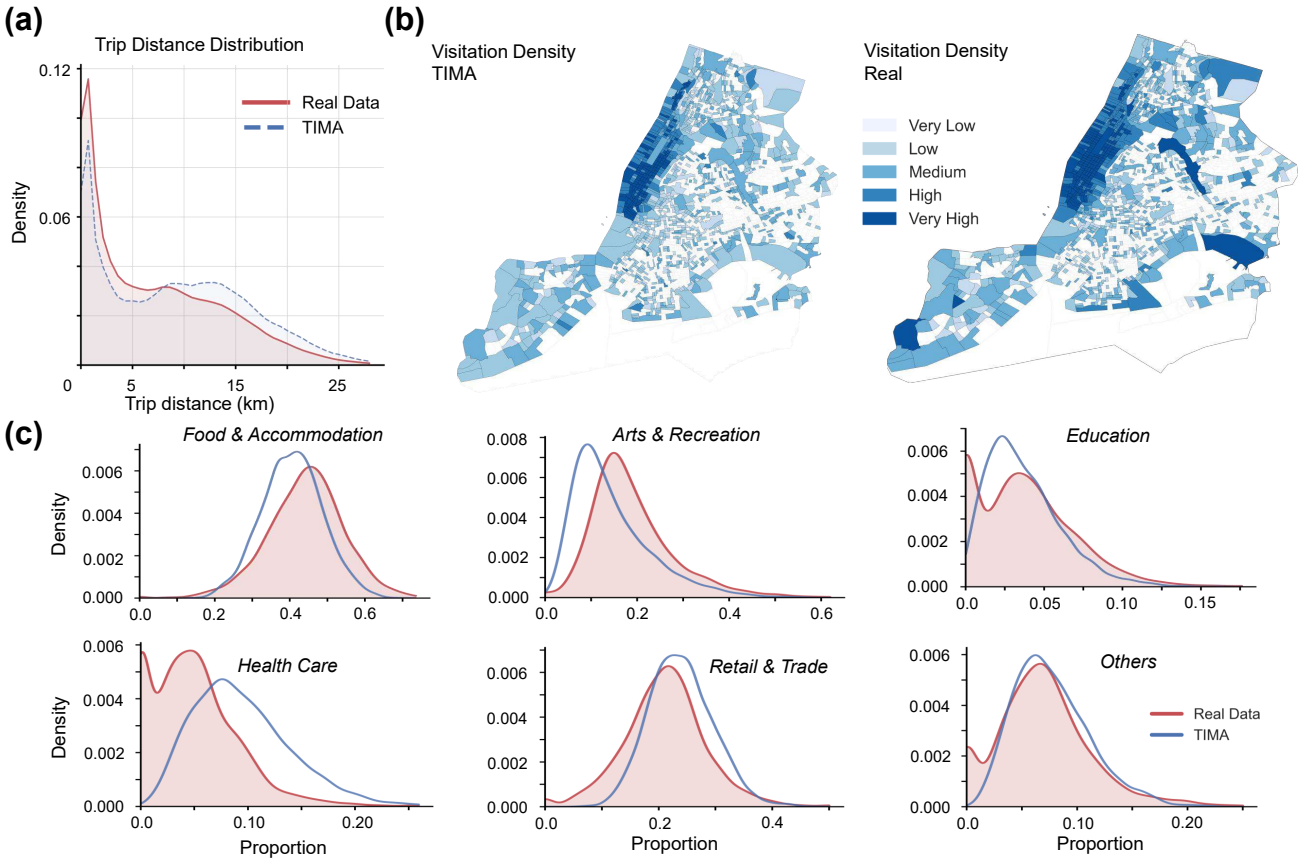


Figure 2. Reproduction of multi-scale mobility patterns in New York City. a, Comparison of trip distance distribution (in km). The simulated distribution (blue) closely follows the heavy-tailed decay (red) of empirical data, capturing the characteristics of Lévy flight[7]. **b,** Comparison of visitation density map aggregated at the Census Tract level. The color scale (very low, low, median, high, very high) is defined by Fisher-Jenks natural breaks[36] ($k = 5$) according to the visitation-density distribution across locations. **c,** Comparison of the proportion of visits by different POI categories at the CBG level between TIMA (blue) and empirical data (red). The overlapping curves indicate that the TIMA framework reproduces the functional demand profiles of different neighborhoods, with quantitative agreement measured by the KL divergence as reported in Table 1.

We evaluate TIMA against four theory-driven baselines across four US cities using observed mobility data averaged over the weeks of June 2019, a pre-pandemic reference period that captures general human movements (see Table 1). Robustness analyses indicate that the relative performance is stable across alternative evaluation windows and seasons; detailed results are reported in Supplementary Information Section 3.6. We assess performance using five complementary metrics that evaluate population-level mobility patterns (trip distances, aggregate OD flows, and visitation density) and behavioral heterogeneity (proportion of visited POIs and stratified OD fidelity). The simulation timesteps are set to 50 steps per week, which approximates the average weekly trip frequency of urban residents reported in empirical studies[37]. We compare against baselines that capture major modeling assumptions the field: 1) the Gravity Model[1], driven by population mass and distance; 2) the EPR Model[10], which introduces exploration and preferential return mechanisms; 3) the d-EPR Model[11], which refines behavior by constraining exploration with gravitational pull and distance decay; and 4) the Social-EPR Model[12], which further integrates the dimension of social stratification. Detailed baseline implementations are described in the Supplementary Information Section 2.9. We explicitly exclude supervised deep learning models (e.g., DeepGravity[38]) as they require training on ground-truth mobility data. We also exclude pure LLM-based agents due to computational constraints that limit them to small-scale simulations ($<$ 1000 agents), preventing the observation of the emergent population-level phenomena targeted here. Importantly, TIMA scales by inferring behaviors once per unique demographic profile. Across four cities ($\sim$3,000 profiles), the total LLM inference cost is around $200, enabling large-scale simulation at modest cost without fine-tuning (see Supplementary Information Section 1.3).

TIMA robustly reproduces population-level mobility patterns across cities. First, it accurately recovers the empirical trip distance distribution, with distances calculated as the Haversine distance[39] from each agent's home CBG to their visited locations. TIMA achieves the lowest trip distance KL divergence in all four cities (NYC 0.054; Chicago 0.007; Orlando 0.042; Norfolk–Virginia Beach 0.049; Table 1) and the simulated distance distribution closely follows the empirical heavy-tailed nature of human mobility, often described as Lévy flights scaling[7] (see Fig. 2a for an example of New York, additional cities and baselines are reported in Supplementary Information Section 3.1). Second, TIMA generates realistic CBG-to-CBG OD structures, as shown in Table 1. The OD alignments are quantified using the Commuter Public Component (CPC)[40], a widely used classic metric in mobility prediction that measures the overlap between two flow distributions. OD flow overlap improves in New York City and remains close to the strongest physics-constrained baseline (d-EPR) in the other three cities (e.g., Chicago 0.384 vs 0.402; Orlando 0.550 vs 0.557; Norfolk–Virginia Beach 0.522 vs 0.550). Finally, visitation density patterns align well with empirical hotspots (Fig. 2b), with TIMA yielding lower density error than all baselines in three of four cities at the CBG level (the exception is Chicago; Table 1).

Beyond matching population-level movement structure, TIMA also captures heterogeneity in both destination choice and group-specific mobility structure. We quantify trip purpose heterogeneity by calculating the proportion of visits assigned to six POI categories and comparing the distributions of these proportions between simulation and observed data. The six POI categories, determined from NAICS codes, are Food & Accommodation, Arts & Recreation, Education, Health Care, Retail & Trade, and Others (see Methods Section "Data" for details). Using New York City as an example, Fig. 2c reveals that the simulated CBG-level visitation proportions across POI types closely track empirical patterns. Table 1 shows that TIMA substantially reduces POI-proportion KL in the largest metros, achieving 0.181 in New York City (76-87% improvement over the baselines) and 0.207 in Chicago (64-78% improvement over the baselines). We further test whether the model preserves group-specific mobility structure using stratified OD fidelity, defined as the mean CPC overlap of flows computed across socioeconomic strata (see Methods section "Evaluation protocol" for details). TIMA achieves the highest stratified fidelity in New York City (0.231) and Orlando (0.462), and remains close to the strongest baseline in other cities (Table 1). Together, these results reveal that TIMA can recover not only aggregate mobility statistics but also meaningful heterogeneity in where people go and how mobility structures vary across diverse subgroups, with the largest benefits emerging in large, heterogeneous urban centers.

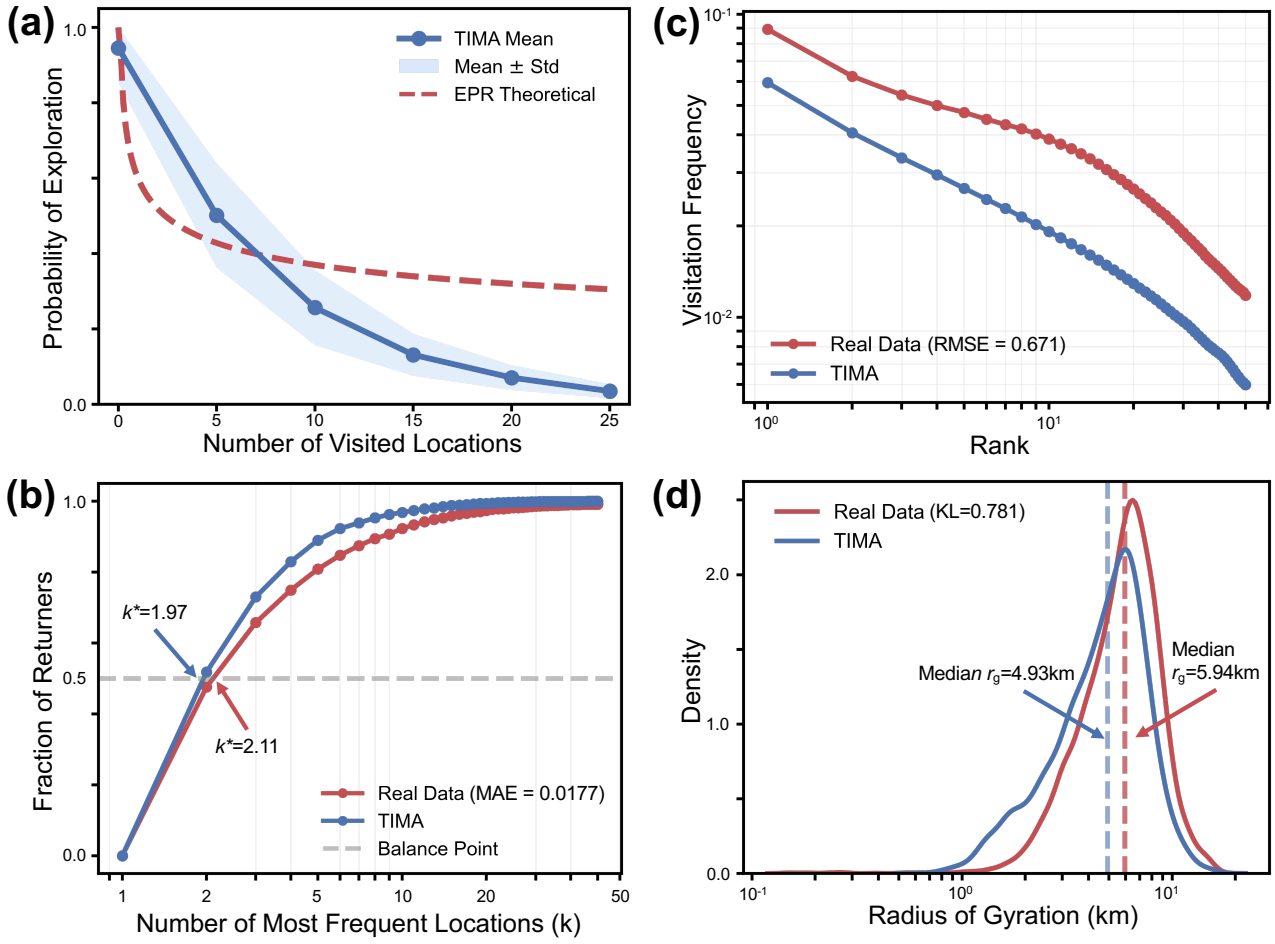


Figure 3. Emergence of fundamental laws and adaptive mobility mechanisms. a, Personalized probability of exploration: comparison of agent-specific exploration tendencies against (Blue) the EPR reference curve (red), where the blue band (mean ± standard deviation) summarizes variation across agents inferred by TIMA from agents' profiles. **b,** Explorers–returners dichotomy: fraction of returners as a function of k, where returners are dominated by their top-k most visited locations. The gray dashed line indicates the population balance point. **c,** Zipf-like visitation scaling: The relationship between visit frequency and ranking in a log-log scale. **d,** Movement radius: probability density of the radius of gyration (r_g). For sub-figures **b** to **d**, blue represents the results from TIMA and red represents observed data.

LLM-driven heterogeneity recovers core mobility regularities

In what follows, we use New York City as an example; similar results for the other cities and baseline models are reported in Supplementary Information Section 3.4. In TIMA, the explore–return decision is modeled as an individualized, history-conditioned choice generated by the LLM. We pre-generate exploratory decision traces for each agent and fit an agent-specific exploration function of unique locations visited (Fig. 3a). Instead of following a universal fixed decay law ($P_{new} \propto S^{-\gamma}$, the blue dashed line) from the EPR model[10], TIMA yields a heterogeneous set of explore–return dynamics. Specifically, exploration declines as the number of visited locations increases, but with substantial between-agent variation, visualized by the agents' mean and dispersion (blue curve and shaded band; Fig. 3a). In light of this, we classify agents as returners if their mobility is dominated by their k most frequently visited locations, and explorers otherwise[11] (see Methods section "Evaluation protocol" for details). As the number

of unique locations visited k increases, individuals are more likely to establish routines and transition from explorers to returners. (Fig. 3b reveals that TIMA closely reproduces the empirical returner fraction as a function of k (MAE $= 0.0177$, calculated as the average absolute difference between the two distributions). To further validate this, we identified the transition point (k^*) where the population reaches a balance between explorers and returners. TIMA yields $k^* = 1.97$, closely consistent with the empirical value of 2.11. Together, those results reveal that the TIMA replaces a single, universal exploration law with individualized explore–return dynamics that remain consistent with population-level regularities.

We next examine whether TIMA follows Zipf's law of human mobility in terms of visit frequency. In Fig. 3c, both the empirical trajectories (red) and TIMA simulations (blue) exhibit an approximately linear relationship on log–log axes, confirming Zipf-like decay in visitation frequency with location rank. Quantitatively, for agents who visit at least 5 unique locations, TIMA achieves the lowest error in the rank–frequency distribution (RMSE = 0.671), outperforming the Gravity Model (1.727) and d-EPR Model (1.104) baselines (see Supplementary Information Section 3.4). Furthermore, Fig 3d evaluates heterogeneity in individual mobility range via the radius of gyration, r_g, defined as the root-mean-square distance of visited locations from the trajectory centroid. TIMA recovers the overall shape of the empirical r_g distribution with a KL divergence of 0.781. Specifically, the median of r_g of the TIMA's result is 4.93km, comparable to the empirical median of 5.94km. Together, these results reveal that the TIMA simultaneously preserves the scaling structure of location preferences and the diversity of spatial visiting scopes observed in real data.

Emergence of socioeconomic segregation and lifestyle heterogeneity

While reproducing population-level mobility structures, TIMA simultaneously captures the heterogeneity in destination choices and travel purposes across socioeconomic groups. To demonstrate this, we compute the experienced segregation[41] (see Methods Section "Evaluation protocol" for details), an activity-space index that measures income similarity between a resident's home neighborhood and the neighborhoods of people they are likely to encounter at the places they routinely visit (higher values indicate more income-homophilous exposure). This segregation signal is calculated from simulated mobility choices (who goes where), agents are initialized from CBG-level population profiles, and we stratify results by residential income rank to report the average simulated exposure pattern for individuals living in neighborhoods of a given income level. As shown in Fig. 4c, TIMA reconstructs the spatial distribution of experienced segregation across New York City, with a simulated index of 0.570, close to the empirical observation of 0.584, and improving upon baselines (ranging from 0.522 to 0.546, see Supplementary Table 7). This structural alignment stems from the model's ability to capture income heterogeneity in destination choice. In both real and simulated mobility (Fig. 4b), individuals from higher-income neighborhoods tend to visit higher-income locations, producing a significant positive association between residential income rank and the mean income rank of visited places (Pearson's $r = 0.30$ in real data versus $r = 0.23$ in TIMA). This hemophilic behavior drives the emergence of stratified segregation patterns as demonstrated in Fig. 4a, where low-income individuals experience relatively higher segregation and more limited cross-income contacts. Figure 4e compares statistically significant hot spots visits inferred from real mobility data with those generated by TIMA, shown separately for high- and low-income groups. Hot spots are identified using a Getis-Ord Gi statistic[42] (see Method Section "Hot Spot Analysis" for details), and we visualize statistically significant high-activity hot spots at the $p < 0.05$ level. Under this criterion, TIMA recovers the dominant geographic pattern of hot spots in both strata. Quantitatively, hot spot overlap reaches 70% for high-income and 62% for low-income groups, indicating that TIMA reproduces the spatial footprint of concentrated visiting activity at scale.

To probe the micro-level mechanism behind these macro-patterns, Fig. 4d compares the POI-category visitation profiles aggregated by socio-demographic group in the data and in TIMA. For real data, we pool residents from CBGs dominated by the target profile (e.g., Black and high-income) and compute their category-level visitation distribution; we construct the simulated counterpart by aggregating trips from the generative agents assigned to the same profile. The TIMA successfully reproduces anisotropic lifestyles driven by intersecting demographic attributes. For instance, high-income Black individuals (Case A) show a clear preference for Arts & Recreation, whereas low-income Black individuals (Case B) exhibit a distribution shifted towards Food & Accommodation services

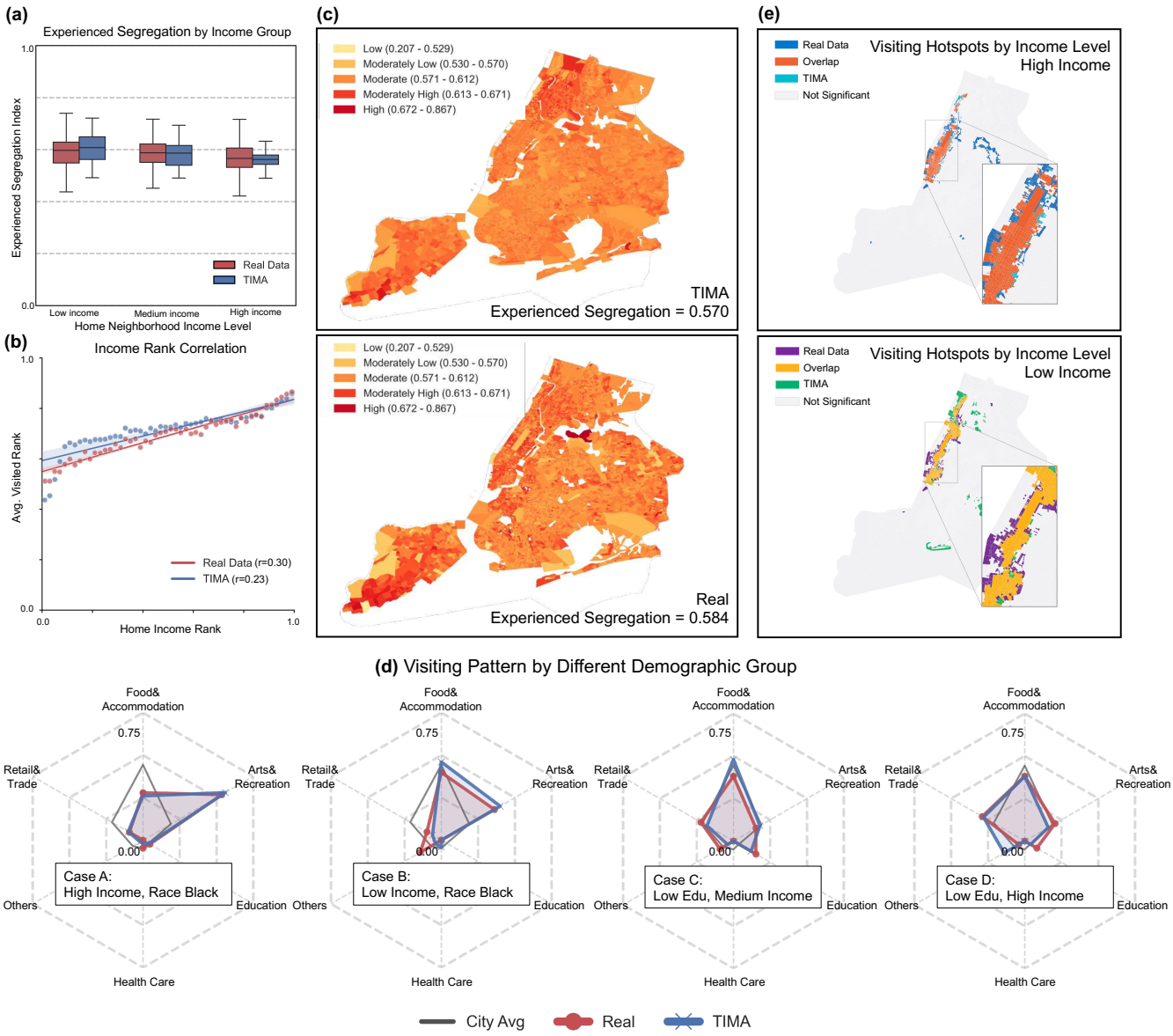


Figure 4. Emergence of socio-economic segregation and lifestyle heterogeneity. a, Experienced segregation by income group. Box plots compare the distribution of experienced segregation indices across three income levels. Red boxes represent real data, blue boxes represent the TIMA's simulation. **b,** Income rank correlation between the income ranking of an individual's home CBG and the average income of their visited locations. **c,** Experienced segregation map in New York City at the CBG level, where darker red indicates higher levels of social segregation. **d,** Anisotropic visiting preferences. Radar plots of visitation probability for four different cases of distinct demographic profiles across six POI categories. The gray line represents the city-wide average of New York City. **e,** Statistically significant high-activity visiting hot spots ($p < 0.05$) are shown for high- and low-income groups, comparing real mobility data (blue/purple) and TIMA simulations (cyan/green). Overlapping hotspots are highlighted (orange/yellow); gray denotes not significant areas. Insets zoom into the Manhattan core.

with reduced visitation to leisure venues, reflecting income-based constraints on lifestyle choices[12,43]. Furthermore, regardless of income level, low-income individuals are consistently depressed about entertainment locations (Cases C and D). Overall, these results indicate that the generative agents in the TIMA effectively infer how socioeconomic and demographic attributes collectively shape heterogeneous mobility behaviors.

Counterfactual pandemic response and emergent segregation

A core uniqueness of the TIMA is its ability to respond to previously unseen policy and crisis scenarios by generating counterfactual mobility patterns through agent-level reasoning. To evaluate this capability, we apply the TIMA to simulate the week following stay-at-home orders in Manhattan, New York, at the beginning of the COVID-19 pandemic. In this experiment, agents infer a personalized distance decay exponent conditioning on individual demographics and external situational context, including public health policy and local infection risks (see Supplementary Information Section 2.8). To prevent the LLM from retrieving specific historical COVID-19 records from its pre-training corpus, we anonymized the virus name and the simulation's temporal identifiers in the prompts. This design ensures that the resulting mobility patterns reflect behavioral adaptation to unseen policy and risk conditions without extra training.

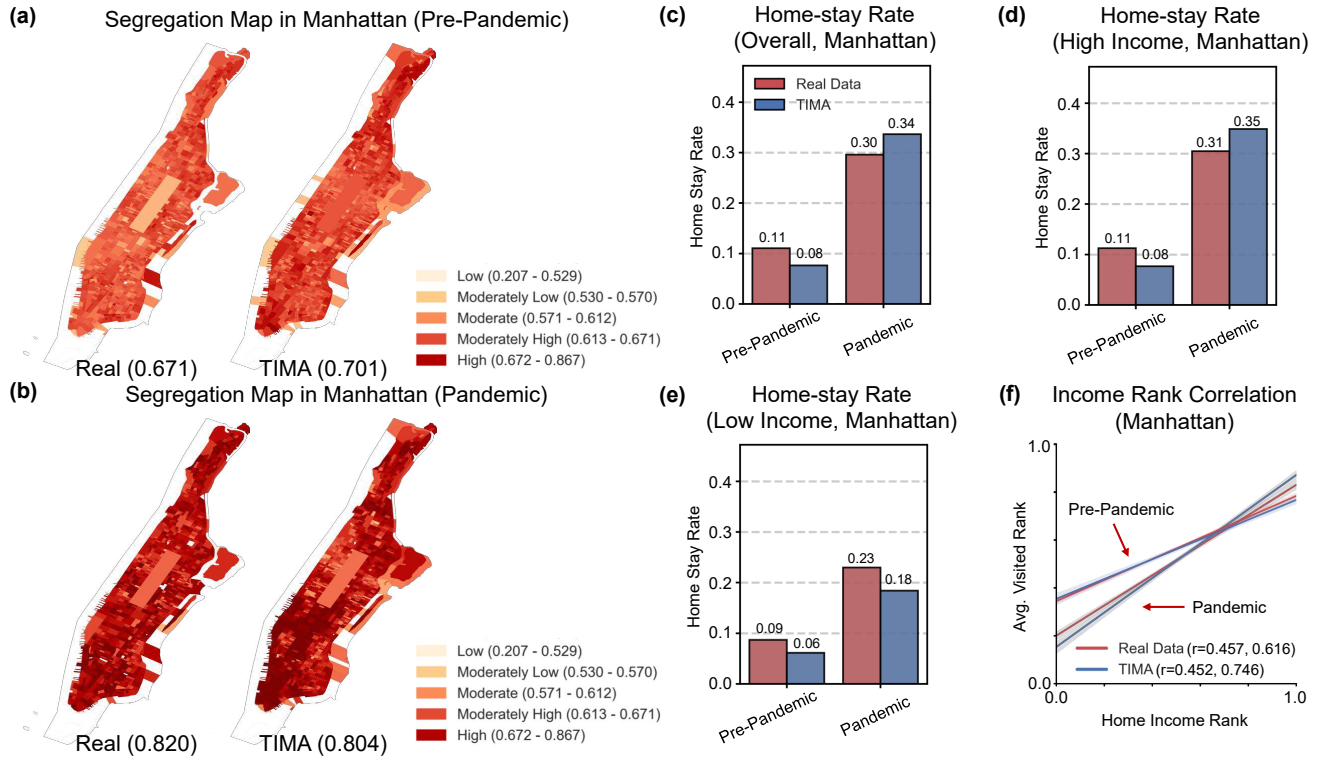


Figure 5. Socioeconomic disparities in home staying and structural segregation during the pandemic context. **a,b,** Spatial distribution of experienced segregation indices at the CBG level for pre-pandemic **(a)** and pandemic **(b)** periods. The index values are provided in parentheses; darker red indicates higher levels of social segregation. **c-e,** Comparison of the home-stay rate, measured as the proportion of individuals whose visits were entirely confined within their home neighborhood's CBG. Panels show the aggregate population response **(c)** and differences among social classes, including high-income **(d)** and low-income **(e)** groups. **f,** Correlation between home income rank and visited location income rank for pre-pandemic and pandemic periods. The steeper slope of the regression line during the pandemic (lower lines) compared to the pre-pandemic period (upper lines) indicates intensified income homophily and social segregation.

Home-stay behavior was widely used as a practical proxy for social distancing and policy compliance during the COVID-19 Pandemic[44,45]. Accordingly, we quantify home-stay using a mobility-based proxy defined as the proportion of individuals whose observed (or simulated) visits are entirely confined within their home neighborhood's CBG. As shown in Fig. 5c−e, the TIMA successfully reproduces the sharp increase in home-stay behavior observed since the stay-at-home orders. In Manhattan, the simulated home-stay rate rose significantly from the pre-pandemic baseline to 0.34, within 4% of the empirical data of 0.30 (Fig. 5c). Beyond matching the population-level shift

to higher home-stay rates, TIMA also captures the pandemic's inequitable burden on mobility reductions across socioeconomic groups. Empirical data reveal that the home-stay rate varied notably by income level: in Manhattan, high-income individuals exhibit the highest surge (Fig. 5d, from 0.11 to 0.31), whereas low-income groups showed a more constrained response (Fig. 5e, from 0.09 to 0.23). The TIMA framework accurately reconstructs this difference, predicting a pronounced shift for high-income individuals (to 0.35) while identifying the sustained, relatively high mobility demands for low-income individuals (to 0.18). This pattern aligns with prior studies[45,46] that high-income groups have a stronger capacity for social distancing through remote work and financial buffers, while low-income groups often face necessity-driven travel due to employment or economic constraints.

The unequal mobility contraction propagates to the city scale by reshaping the macroscopic social mixing. We examined this using the experienced segregation index introduced above. As illustrated in Fig. 5 a,b, the pandemic triggered a sharp rise in segregation. In Manhattan, the empirical segregation index increased from 0.671 to 0.820, and TIMA closely reproduces this trend (0.701 to 0.804), while capturing its spatial redistribution across CBGs. Consistent with this segregation increase, Fig. 5f shows a strengthened coupling between mobility and home CBG income. As mobility became restricted, the correlation between residents' mobility patterns and the income level of their home neighborhoods became stronger. Empirically, the correlation rose from 0.457 to 0.616, and TIMA predicts a comparable strengthening (0.452 to 0.746). These results suggest that TIMA can be used prospectively to anticipate real-time mobility responses to future pandemics from policy and risk cues, without requiring outbreak-specific mobility data.

Discussion

In this study, we propose TIMA as a hybrid framework for human mobility modeling that integrates physical mobility theories with generative intelligence. Physical theories provide a strong inductive basis, but typically represent only a limited subset of mobility dimensions and flatten individual heterogeneity[31]. Generative AI, in contrast, can express rich individual variation but often lacks guarantees of physical plausibility. TIMA combines the two: mobility theory constrains what must hold at scale, while LLM-driven agents generate personalized, context-sensitive decisions conditioned on agent persona and local environments. This division of effort preserves population-level regularities while producing interpretable, agent-specific drivers of behavior (e.g., personalized exploration of return shown in Fig.3a and heterogeneous lifestyles shown in Fig.4d). The result is a data-light simulator that reproduces both realistic individual mobility patterns and their social consequences, offering a practical mobility proxy when fine-grained trajectory data are unavailable or access-restricted.

Across multiple cities and evaluation metrics, TIMA reproduces heavy-tailed travel distances and coherent OD structure, while also improving fidelity in destination-type composition and stratified flows. Notably, the largest performance gains emerge in heterogeneous metropolitan areas, suggesting that performance may benefit from the richer behavioral and contextual diversity represented in the data used to pretrain modern LLMs[47]. Beyond population-level fit, TIMA offers a mechanism-grounded account of how mobility regularities can be preserved while allowing individualized variation. Instead of imposing a single exploration law, TIMA generates history-conditioned exploration propensities that vary across agents (Fig. 3a). Notably, the emergent decay toward near-zero exploration as the visited set approaches around 25 locations mechanistically reproduces prior evidence that individuals maintain a conserved, finite-capacity activity space—typically limited to around 25 familiar locations at any given time[48].

A central scientific contribution is that TIMA can be used not only to simulate mobility but also to study social phenomena produced by mobility as a side effect, such as social segregation. This aligns with previous work showing that human mobility data can quantify experienced urban inequalities, in realized exposure and access shaped by where people actually go[49]. By explicitly incorporating individual heterogeneity in preferences and constraints, the framework provides a micro-to-macro account of how destination choices translate into collective social structure, exposing a Schelling-style segregation mechanism in activity space: small, individual-level biases in where people go can accumulate into city-scale segregation in experienced contact[50]. More importantly, TIMA shows that this structure is dynamic under external shocks (Fig. 5): during the pandemic, unequal mobility options reconfigured social mixing by shrinking activity spaces and tightening the coupling between mobility and income, amplifying

homophilous exposure and increasing experienced segregation[51]. Ultimately, TIMA advances a generative approach to understanding human society, moving beyond the simulation of mobility patterns.

Most importantly, TIMA enables prospective counterfactual simulation of behavioral response during crises and rapid policy shifts. During the past pandemic, when behavior changes fastest and interventions are most consequential, fine-grained mobility observations are often delayed or incomplete[52,53], limiting the ability to anticipate how people will actually respond. As a result, decision-makers struggled to evaluate intervention effectiveness prospectively and to anticipate downstream outbreak dynamics and consequences[54,55]. TIMA addresses this gap by coupling mechanistic mobility constraints with LLM-driven agents that can reason over unstructured, rapidly evolving scenarios (e.g., local disease risk information and policy directives) and translate them into adaptive, interpretable parameters for simulation of counterfactual human responses. Moreover, the simulated, scenario-based mobility trajectories and contact proxies produced by TIMA can be directly integrated as dynamic inputs to infectious disease transmission models[56], enabling human–disease interaction modeling under counterfactual interventions. This flexibility allows the same simulation framework generalize across different external shocks, whether infection risk, closures, reopens, or targeted restrictions, without requiring retraining on task-specific data. As a result, TIMA can serve as a practical decision-support tool, enabling policymakers to test intervention designs and evaluate policy trade-offs before consequence data become available. More broadly, this prospective capability generalizes beyond pandemic response to other disasters and disruptions (e.g., wildfire, hurricanes, and evacuation orders), providing a unified way to anticipate population response under shifting risk and policy conditions.

This study has several limitations. First, although we benchmark against SafeGraph mobility data, these observations remain a subset sample of real-world individuals. Consequently, inferred mobility patterns may be systematically biased for populations with lower smartphone ownership (e.g., children) and for minority communities that are underrepresented in the panel[57]. To address this, we report coverage-aware validation, stratifying performance by CBG device coverage (see Supplementary Information Section 3.3); performance maintains as data coverage increases. The framework would benefit from recalibration as higher coverage or more representative mobility data become available. Second, several dimensions of human mobility are not explicitly modeled in this study, such as time-of-day effects and activity duration. Omitting these temporal components can limit realism in the activity trajectory. The TIMA framework is intentionally designed to be flexible and can be extended to incorporate these dimensions. For example, trip duration can be incorporated into the physical backbone via empirically calibrated stochastic sampling[58], while time-of-day structure can be incorporated into the LLM layer as explicit conditioning variables (e.g., commuting windows and work commitments)[16]. A further limitation is that the LLM introduces variability that can potentially affect the preferences inferred from agents. Agent decisions may shift with prompt phrasing or decoding settings (e.g., temperature), and these shifts can propagate to downstream mobility patterns. In addition, different LLM architectures may exhibit model-specific biases, leading to agents that encode demographically stereotyped[59]. To reduce these risks, we standardize prompts and decoding configurations, evaluate robustness under different temperature settings within the same LLM (Supplementary Information Section 3.7), and replicate the full pipeline across three distinct LLM architectures (Supplementary Information Section 3.8). Across temperature settings (0.0–1.0), all macro-level mobility metrics remain effectively unchanged, indicating negligible sensitivity to stochastic decoding. Moreover, performance remains largely consistent across Gemini-2.5-Pro, GPT-4o-mini, and Qwen-2.5-7B-Instruct, demonstrating that the TIMA framework is model-agnostic and that its system-level behavior is not dependent on a specific LLM implementation. Looking forward, we will incorporate stronger behavioral regularization (e.g., group-specific fine-tuning of agents) to improve the stability and reliability of the inferred decision rules.

While subject to limitations, TIMA sheds light on the potential of hybrid, theory-guided generative frameworks for modeling human mobility. By conditioning on theory-grounded mobility constraints and context-dependent LLM inference, we show that TIMA can reproduce patterns of human mobility at scale, offering a bridge between human behaviors and computational modeling. Future work can build on this foundation by extending it to other settings and by validating agent-level preferences through targeted data collection. We envision a future in which such AI technologies enhance our understanding, anticipation, and response to larger-scale behavioral change.

Methods

In this work, we introduce the Theory-Informed Mobility Agents (TIMA) framework, which integrates classical physical theories with generative semantic reasoning to simulate human mobility. Agents are instantiated from publicly available demographic and socioeconomic information and move within a digital environment defined by POI categories and CBG boundaries. Agents' actions are governed by an exploration–return decision structure grounded in EPR theory. Building upon this structure, TIMA utilizes context-dependent reasoning to dynamically derive personalized trip purpose and destination preferences through LLMs, conditioning on individual demographic and socioeconomic profiles, local urban semantics, and external situational context. Crucially, TIMA operates independently of empirical mobility data; raw mobility data are never used to train, fit, or calibrate the simulator, but are used only as external ground truth for evaluation. The mobility data is obtained from SafeGraph, a data company that aggregates anonymized location data from numerous applications in order to provide insights about physical places, via the SafeGraph Community. To enhance privacy, SafeGraph excludes census block group information if fewer than two devices visited an establishment in a month from a given census block group. The TIMA framework produces individual mobility trajectories, which are then aggregated to assess TIMA's capacity to recover population-level flow patterns and segregated social stratification structures across multiple U.S. metropolitan areas of varying scales and morphologies.

Data

Our study integrates three categories of data: 1) structural inputs for agent instantiation and action guidance, 2) situational context information for counterfactual reasoning, and 3) empirical human mobility data for evaluation.

Structural inputs for agent simulation. These datasets provide the information accessible to generative agents during simulation and comprise two complementary components: a) persona data, which include the demographic and socioeconomic attributes used to construct heterogeneous agent profiles, and b) environmental data, which capture the physical layout and semantic structure of the local environment that shapes agents' decision-making.

- **Demographic attributes.** We obtain CBG–level demographic and socioeconomic data from the U.S. Census Bureau's American Community Survey (ACS)[60], including race, sex, age, and occupation. The demographic information utilized in this project is derived from the 2019 5-year ACS.

- **Socioeconomic indicators.** We obtained CBG level socioeconomic indicators, including household income and educational level, from the urban sustainability dataset[61]. These socioeconomic indicators were combined with ACS demographic distributions to jointly initialize heterogeneous agent profiles.

- **Local environments.** We obtain POI data from the SafeGraph[32]. Each POI is associated with precise geographic coordinates and a category code defined by the North American Industry Classification System (NAICS)[62]. These categorical labels define the activity space available to agents during simulation. To construct the agent's semantic state space, we aggregate the 20 primary NAICS sectors into 6 core functional categories: a) Food & Accommodation, b) Arts & Recreation, c) Education, d) Health Care, e) Retail & Trade, and f) Others (see Supplementary Table 1 for detailed mapping corresponding to the NAICS code).

- **Geographic boundaries of CBGs.** Census standardized geographic shapefiles defining CBG boundaries are used to spatially reference agents and POIs[63]. By mapping POIs to the geographic boundaries, we construct a continuous urban semantic space that enables agents to perform decision-making reasoning based on spatial awareness and semantic context.

Human mobility data. To obtain real-world mobility patterns as a benchmark, we employ the SafeGraph Weekly Patterns dataset[32]. Our evaluation spans four U.S. metropolitan areas are selected based on varying city scales and morphologies: New York City, which is a high-density metropolis characterized by heavily relying on public transportation and complex functional urban landscape; Chicago, which is another larger but geographically distinct city with a smaller amount of population and urban scale than New York City; Orlando, which is a tourism-oriented

city in the South; and Norfolk-Virginia Beach, representing a typical small city in the U.S. This dataset is derived from aggregated anonymous location data from millions of mobile devices, and reports aggregated visit counts from CBG to different categories of POIs.

We construct two benchmarks for two different periods to evaluate TIMA's performance under both normal and crisis conditions: 1) Normal routine mobility benchmark: We use data from the weekly average of June 2019, which is randomly selected as a representative of pre-pandemic travel patterns. Results for different periods are provided in Supplementary Information Section 3.6. 2) Crisis response benchmark: We use data from the week immediately following the issuance of stay-at-home orders in early 2020 to validate the model's adaptability to emergent situations. We extracted fine-grained origin-destination (OD) flows from this data, quantifying the number of visits from home CBG i to POI j. These mobility data were used exclusively as external ground truth for evaluation and were not accessible to agents or used to parameterize the simulation. Detailed data processing steps are documented in the Supplementary Information Section 1.4.

Situational context data. To support counterfactual analysis of crisis responses, we compiled pandemic situational context from two official sources: (i) epidemiological surveillance from the JHU CSSE COVID-19 dashboard[64], including cumulative reported cases and deaths; and (ii) public health policies and timing from the Oxford COVID-19 Government Response Tracker[65], including the timing and content of state stay-at-home orders, recommended social distancing protocols (e.g., maintaining 6 feet of distance), and category-specific operating restrictions (e.g., bars closed; restaurants limited to takeout; non-essential business closures). We synthesized these materials into a concise textual scenario used as input to the agents' reasoning. See Supplementary Information Section 1.5 for data details.

The Theory-Informed Mobility Agents framework

The proposed Theory-Informed Mobility Agents (TIMA) framework bridges physical mobility theory with microscopic LLM-inferred personalized preferences, as shown in Fig. 6b. TIMA preserves the universal regularities of human movements via exploration and preferential return (EPR) theory[11], while using LLM reasoning to infer the agent-specific parameters that instantiate these mechanisms. Representative agents are initialized using CBG-level demographic and socioeconomic attributes and move through a spatially and operationally feasible urban opportunity set. During the simulation, agents follow a two-stage decision process: 1) a personalized history-dependent exploration-preferential return decision, and 2) a semantic–gravity destination selection conditioned on theory-informed constraints. This hybrid framework allows individual heterogeneity to emerge under grounded in the universal scaling laws. A summary of the TIMA design is presented below, with full methodological details provided in the Supplementary Information Section 2.

Agent initialization. To represent the demographic heterogeneity at the individual level, we adopt the Iterative Proportional Fitting (IPF)[66] to construct the representative generative agents. For each CBG i, agents are initialized with a persona vector $\mathbf{v}_u$ with demographic and socio-economic attributes as demonstrated in Fig. 6a, including the agent's sex, age, race, and home neighborhood CBG's income and education level. These attributes are sampled from joint distributions matched to the agent's home neighborhood CBG's census marginals. We instantiated $n = 10$ agents per CBG to balance computational tractability with statistical representativeness, resulting in 64930 agents in New York City, 21830 in Chicago, 1440 in Orlando, and 4910 in Norfolk-Virginia Beach. In addition to the demographic information, we also assigned a home environmental context vector $\mathcal{H}_u$ to each agent, aiming to represent the local functional composition and serves as a proxy for the residential environment, thereby endowing the agent with basic environmental awareness and ensuring that the generated preferences are grounded in its residential reality (Fig. 6a). The detailed procedures for attribute discretization and the construction of $\mathcal{H}_u$ are provided in Supplementary Information Section 2.1. Together, we denote by $\Theta_u = \{\mathbf{v}_u, \mathcal{H}_u\}$ the complete set of agent-specific information used to parameterize agent u throughout the simulation.

Exploration and preferential return. At each time step t, agent u visits a specific location l_u^t, where l_u^t is associated with a POI category $k(l_u^t)$ and a CBG $c(l_u^t)$. Each agent also maintains an ordered visitation history $\mathcal{L}_u^t = \{l_u^0, ..., l_u^t\}$ that contains the set of all locations visited up to the moment t. Based on this history, the agent first determines

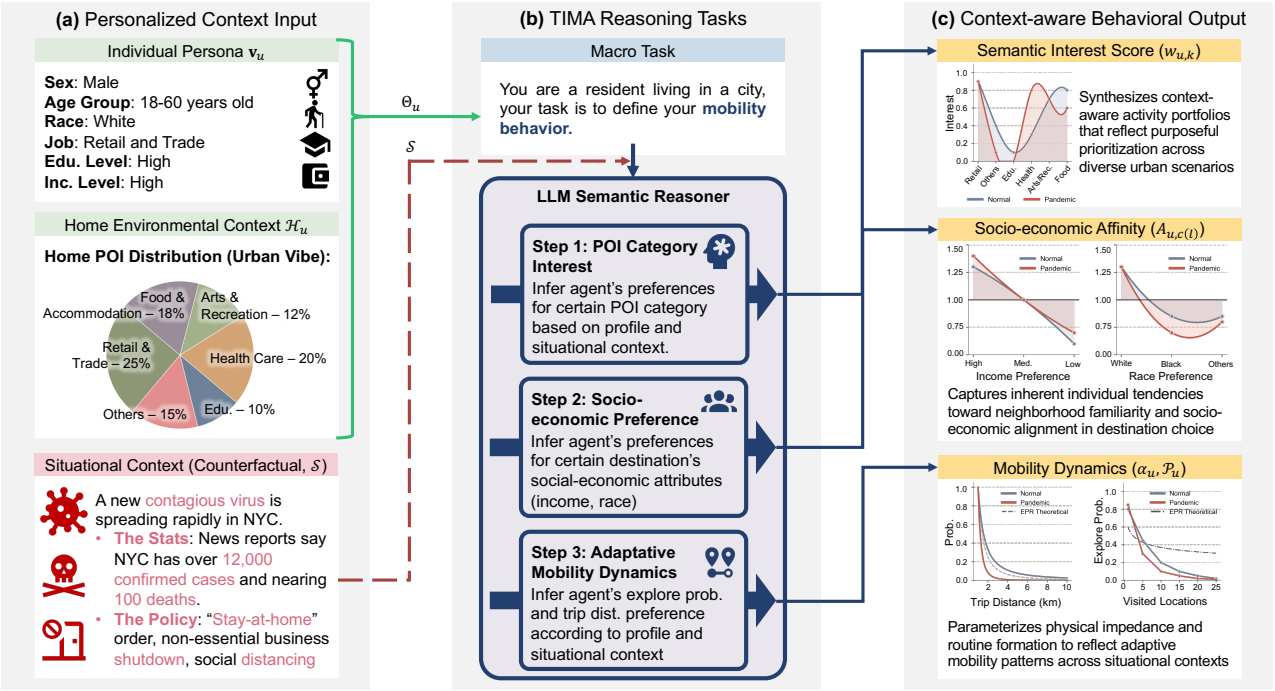


Figure 6. Modular prompt design and reasoning workflow of TIMA framework. a, Personalized context input. The TIMA framework synthesizes individual-level prompts from three primary sources: agent persona $\mathbf{v}_u$, detailing demographic and socio-economic attributes; home environmental context $\mathcal{H}_u$, representing the home neighborhood CBG's POI distribution and severing as a proxy of residential environment; and situational context $\mathcal{S}$, incorporating external crisis and policy requirements for counterfactual analysis. **b,** TIMA reasoning tasks. The LLM reasoner processes the unified context input through a structured three-step pipeline, and TIMA then further identifies the behavioral logic of the agent's urban functional (POI category) interest, socio-economic neighborhood preference, and the adaptation of mobility patterns to situational constraints. **c,** Context-aware behavioral output. The reasoning outcomes are manifested as context-specific parameters modulating the physical driver of human mobility: semantic interest score $w_{u,k}$, socio-economic affinity weights $A_{u,c(l)}$ and mobility dynamics parameters (α_u and $\mathcal{P}_u$). The visualization compares the differences between a normal routine and a pandemic-counterfactual scenario.

whether to explore a new location or return to a previously visited one, following the fundamental structure of the EPR model. As in classical EPR models, this decision depends on the agent's number of distinct locations previously visited and follows a fixed decay law. However, previous studies suggest that the mobility patterns are highly diverse[67] and dependent on individual demographic and socioeconomic backgrounds[12]. Therefore, TIMA allows the decision of exploration to be shaped by heterogeneous, agent-specific exploration tendencies as shown in Fig. 6c. Formally, the probability $P^t_{u,\mathrm{e}}$ that agent u explores a new location at time t is modeled as:

$$P^t_{u,\mathrm{e}} = \mathcal{P}_u(|S^t_u| \mid \Theta_u), \quad S^t_u = \mathrm{Unique}(\mathcal{L}^t_u) \tag{1}$$

where the S^t_u represents the set of distinct locations the agent u visited at the time t. Crucially, $\mathcal{P}_u(\cdot)$ is a personalized exploration profile inferred by the LLM conditioned on the agent's profile Θ_u. Rather than using fixed parameters, TIMA defines a dynamic exploration probability based on discrete ranges of previously visited history. This design preserves the core EPR intuition while enabling heterogeneous exploration tendencies across agents with different profiles (see Supplementary Information Section 2.4 for detailed threshold settings).

If not exploring, the agent u enters the preferential return process with probability $P^t_{u,\mathrm{r}} = 1 - P^t_{u,\mathrm{e}}$, and will select destination j from its unique destination history S^t_u to revisit based on probability $P^t_u(l)$, defined as:

$$P^t_u(l^t_u = l \mid \text{return}, \mathcal{L}^t_u) = \frac{f^t_{u,l}}{\sum_{l' \in S^t_u} f^t_{u,l'}}, \quad l \in S^t_u, \tag{2}$$

where $f^t_{u,j}$ denotes the cumulative number of visits by agent u to location l up to time t.

Semantic destination selection during exploration. When an agent decides to explore, the destination choice should reflect both heterogeneous behavioral intent and the environment's external constraints. While traditional mobility models typically represent exploration destinations as a function of distance decay and location assortativity, TIMA extends this by modeling destination choice as a structured human-like decision process. People typically first decide what they want to do (trip purpose) and only then decide where to do it (exact destination), given what is feasible from their current position. Accordingly, destination selection proceeds in two stages (see Supplementary Information Section 2.5 for detailed mathematical formulations). First, the agent chooses a POI category k that encodes its activity purpose, guided by agent-specific semantic preferences and local availability (Stage 1). Second, condition on k, the agent selects a specific location l on a city-wide global scale (Stage 2), integrating destination preference with spatial constraint. This two-stage choice follows the factorization below:

$$P^t_u\big(l^t_u = l \mid l^{t-1}_u, \text{explore}, \Theta_u\big) = \underbrace{P\big(k^t_u = k \mid l^{t-1}_u, \Theta_u\big)}_{\text{POI-category preference}} \cdot \overbrace{P\big(l \mid k(l) = k, l^{t-1}_u, \Theta_u\big)}^{\text{CBG socioeconomic preference}}. \tag{3}$$

Each agent u is associated with semantic purpose-related interest scores $\{w_{u,k}\}_{k=1}^K$, where $w_{u,k} \in [0,1]$ captures the agent's preference for category k. As shown in Fig. 6c, TIMA uses an LLM to infer these preferences by conditioning on agent-specific information Θ_u together with a natural-language description of each POI category; this yields an agent–category preference score $w_{u,k}$ for every agent-POI pair. Crucially, to prevent the LLM from replying on memorized geographical knowledge, the city name is explicitly omitted from the prompt (see Supplementary Information Section 2.3 for full prompt input). To assess the stability of these inferred preferences, we conduct sensitivity analyses by (i) repeating the inference across multiple temperature settings and independent stochastic runs (see Supplementary Information Section 3.7), and (ii) re-running inference under 10 prompt permutations that vary the phrasing of task instructions and the structured presentation of agent profiles, and we quantify the resulting variability in $w_{u,k}$ (see Supplementary Information Section 3.9). Category choice is restricted to locally available opportunities. Let $\Omega_k(l^{t-1}_u, D_{\max})$ denote the set of feasible POIs of category k within distance $D_{\max}$ from the current location l^{t-1}_u (see Supplementary Information Section 2.2 for detailed theoretical derivation for $D_{\max}$). We sample the activity purpose as:

$$P\big(k^t_u = k \mid l^{t-1}_u, \Theta_u\big) \propto w_{u,k} \mathbf{1}\big[|\Omega_k(l^{t-1}_u, D_{\max})| > 0\big]. \tag{4}$$

Conditional on the selected purpose k^t_u, the agent draws a specific POI l from the city-wide candidate set $\mathcal{L}_k = \{l \in \mathcal{L} : k(l) = k\}$ using a semantic–gravity kernel:

$$P\big(l \mid k^t_u = k, l^{t-1}_u, \Theta_u\big) \propto A_{u,c(l)}\, d(l^{t-1}_u, l)^{-\alpha_u}, \quad l \in \mathcal{L}_k. \tag{5}$$

Here $d(\cdot,\cdot)$ is the Haversine distance[39] between POIs, $\alpha_u > 0$ controls distance decay, and $A_{u,c(l)} > 0$ is an agent–CBG affinity weight that captures agent u's destination preference for the socioeconomic environment of the CBG $c(l)$ (Fig. 6c). In TIMA, the LLM infers a distinct affinity value for each agent–CBG pair by conditioning

on the agent's profile Θ_u together with a semantic description of the destination CBG, including its demographic composition and socioeconomic status; this produces $A_{u,c}$ values that explicitly encode where different agents are more (or less) inclined to go. Similar to the inference of $w_{u,k}$, we assess robustness by repeating the affinity inference across multiple temperature settings and prompt settings, and quantify variability in the resulting $A_{u,c}$ (see Supplementary Information Sections 3.7 and 3.9). In baseline simulations, we fix $\alpha_u = 2.0$ to align with commonly used gravity scaling and to enable direct comparison with EPR-family baselines. In counterfactual settings, we additionally allow the LLM to adapt α_u as a function of the situational context (see Method 'Counterfactual simulation setup' Section). To further verify the robustness of the inferred parameters across demographics, we select 10 agents spanning distinct demographic profiles and repeat LLM inference 50 times per agent, quantifying the resulting variability in Supplementary Information Section 3.2.

Conceptually, the semantic destination selection combined sociological cognition with physical laws. The semantic preference $w_{u,k}$ and social-economic affinity $A_{u,c(l)}$ act as the "cognitive mass" driving attraction, while the personalized distance decay exponent α_u ensures the agent's intentions are grounded in geographical reality. This structure ensures that the TIMA adheres to the universal physical constraint and also accounts for individual heterogeneity. Complete model configuration, parameterization, and implementation details are provided in Supplementary Information Section 2.6.

Counterfactual simulation setup

To evaluate the framework's capacity to simulate behavioral shifts during the external emergency crisis, we introduce the situational context into the workflow of the TIMA framework. In the counterfactual experiments, the generation process of each agent's outputs was augmented by injecting a high-dimensional situational context vector $\mathcal{S}$ into its cognitive prompt.

Situational context construction. We construct a concise situational context that summarizes the evolving crisis conditions faced by agents during counterfactual simulations. For the pandemic case study, we integrate two complementary streams of local scenario-based information: 1) Public health policies, including the presence and timing of stay-at-home orders, recommended social-distancing guidance (e.g., 6-foot distancing), and category-specific operating restrictions that directly affect mobility opportunities (e.g., restaurants limited to takeout; bars closed; non-essential business closures). The policy-related information encodes the compliance and opportunity constraints that govern which activities remain permissible and accessible. 2) Local disease burden, including state-level reported cases and deaths, which serves as proxies of risk exposure. To mitigate data leakage from the LLM's pretraining data and ensure decisions are driven by the provided scenario rather than memorized historical narratives, we anonymize the situational description by masking the specific pathogen name (e.g., "COVID-19") and removing exact calendar dates; agents are instead prompted with an "unknown infectious disease" scenario described only through the provided policy and surveillance summaries. The detailed prompt can be found in the Supplementary Information Section 2.8.

Dynamic behavior adaptation. To model adaptive responses under crisis conditions, we allow the LLM to update each agent's behavioral preferences by jointly conditioning on the agent's profile (Θ_H) together with the situational context. Specifically, for each counterfactual period, we update three components. First, we adapt the agent's exploration tendency by recalibrating its exploration function based on the scenario and the agent's profile. This step aims to capture stratified behavioral responses, such as suppressed discretionary exploration for remote-capable agents, while preserving necessity-driven travel for essential workers. Second, we adapt the agent's activity intent by reweighting its semantic purpose-related interest scores over POI categories ($w_{u,k}$) to reflect necessity and accessibility under policy restrictions. Third, we adapt the agent's distance sensitivity by adjusting the distance-decay exponent (α_u) used in the exploration kernel (see Eq. 5). We anchor this parameter to the baseline value used in normal-period simulations (to preserve comparability with theory-based baselines) and let the LLM infer an agent-specific multiplier that increases or decreases travel friction according to the experienced vulnerability and constraints described in the scenario. Full prompting templates, parameterization details, and examples are provided in the Supplementary Information Section 2.8.

Evaluation protocol

Population-level mobility patterns. To quantify the alignment between the simulated and imperial mobility patterns at the aggregate spatial dimension, we utilized the three following complementary metrics:

- **Trip distance distribution.** We compare the probability density of trip distances d between TIMA, $P_{\text{sim}}(d)$, and data, $P_{\text{real}}(d)$. Discrepancy was summarized by the Kullback–Leibler (KL) divergence:

$$D_{KL}(P_{\text{real}}||P_{\text{sim}}) = \sum_{d} P_{\text{real}}(d) \log \frac{P_{\text{real}}(d)}{P_{\text{sim}}(d)}, \tag{6}$$

- **OD flow similarity.** We aggregate individual trajectories into CBG-to-CBG OD flows. To measure the structural overlap between the simulated OD flow matrix T^{sim} and the empirical matrix T^{real}, we employed the CPC metric, which is a robust metric for evaluating mobility network[40]:

$$\text{CPC} = \frac{2\sum_{i,j} \min(T_{ij}^{\text{real}}, T_{ij}^{\text{sim}})}{\sum_{i,j} T_{ij}^{\text{real}} + \sum_{i,j} T_{ij}^{\text{sim}}}, \tag{7}$$

  where T_{ij} represent the flows from tract i to tract j. The CPC ranges from 0 (no overlap) to 1 (perfect structural consistency).

- **Visitation density.** To evaluate the spatial distribution of visiting hotspots at the CBG level, we define a normalized MSE to handle the heavy-tailed visitation counts while avoiding the error caused by extreme hotspots and outliers (e.g., Times Square in New York City). Here, we use V_l to denote the total number of visits to location l. We first apply a log transform, $\hat{V}_l = \log(1+V_l)$, and then rescale these log-counts by dividing by the city-wide maximum so that the resulting values lie in $[0,1]$. The metric is defined as:

$$\text{MSE}_{\log} = \frac{1}{N_{\text{CBG}}} \sum_{l} (\tilde{V}_l^{\text{real}} - \tilde{V}_l^{\text{sim}})^2, \tag{8}$$

  where N_{CBG} represents the total number of CBGs in the city. A lower log-norm MSE value indicates a more accurate reproduction of the urban mobility pattern.

Heterogeneous destination choice and group-specific mobility patterns. We use the following metrics to evaluate the model's capacity to reproduce the structural heterogeneity of travel across different functional demands and socioeconomic strata:

- **Proportion of visits by POI category.** To evaluate whether the model reproduces (i) diverse trip purposes and (ii) neighborhood-level lifestyle heterogeneity, we compare how visit shares are allocated across functional POI categories for residents of each CBG. For every CBG c and category k (with $K = 6$), we compute $p_{c,k}$, the proportion of visits departing from CBG c and traveling to POI category k. This CBG-by-category profile serves as a compact lifestyle signature, reflecting how different neighborhoods distribute their activity portfolios across purposes. We then summarize city-wide variability by estimating, for each category k, the probability density function $P_k(x)$ using kernel density estimation (KDE). The alignment between simulation and data is quantified by computing the average KL divergence across all POI categories' visitation distribution:

$$D_{KL}^{\text{avg}} = \frac{1}{K} \sum_{k=1}^{K} D_{KL}(P_k^{\text{real}} \parallel P_k^{\text{sim}}). \tag{9}$$

This metric tests whether the model reproduces neighborhood-specific activity portfolios, that is, the distinct patterns in how residents allocate visits across POI categories. The lower values indicate better recovery of real patterns.

- **Stratified OD flows.** To evaluate the reproduction of mobility network within demographic groups, we segment the individual agent along six attributes encoded in the persona vector ($\mathbf{v}_{u.}$): income, education, race, occupation, sex, and age. Income strata are defined by quartiles of the empirical income distribution, whereas other strata follow standard Census reporting categories. For each stratum, we aggregate the trajectories into CBG-to-CBG OD flows and compute CPC between simulated outputs and observed data. We report the mean CPC across strata as a summary of subgroup-level agreement.

Physical mobility law evaluation. To verify whether the model follows the universal scaling laws of human mobility, we verify the model outputs with three widely reported regularities:

- **Law 1: Zipf's law.** We compute the population-level average visitation frequency distribution of the k-th most frequently visited location $P(k) \propto k^{-\zeta}$. Here, $P(k)$ denotes the visitation frequency of the $k-th$ most visited location, and ζ is the scaling exponent that controls the decay rate with rank (large ζ indicates faster decay). The degree of fitting was measured by the root mean square error (RMSE) between simulated curves and the ground truth on a logarithmic scale.

- **Law 2: Radius of gyration.** To characterize the spatial range of individual travel, we calculate the radius of gyration for each agent's trajectory, $r_g = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mathbf{r}_i - \mathbf{r}_{cm})^2}$, where r_i is the coordinate of the i-th visit, r_{cm} is the center of mass of the trajectory. We then use the KL divergence to assess the alignment of the population-level distributions between the simulation and the data.

- **Law 3: The "returners and explorers" dichotomy.** Following the definition by Pappalardo et al.[11], we quantify individuals based on their recurrent mobility structure. We calculate the radius of gyration $r_g^{(k)}$ considering only the top k most frequently visited locations. An individual is defined as a "returner" if $r_g^{(k)} > r_g/2$. We quantified the proportion of returners as a function of k, and used Mean Absolute Error (MAE) to evaluate whether the model successfully reproduced the dichotomy between exploration and return behavior.

Social segregation and crisis response. To assess the model's capability in capturing socioeconomic stratification and behavioral adaptation, we use the following indicators:

- **Experienced segregation.** We adopt the experienced segregation index (S)[41] to quantify an individual's exposure to different socioeconomic groups through mobility. Intuitively, the metric increases when an individual's visited places are dominated by visitors with socioeconomic profiles similar to their own. Formally, this index is defined as:

$$S = 1 - \frac{1}{W_{\text{total}}} \sum_{u} \sum_{l} w_{ul} E_{ul} \tag{10}$$

where w_{ul} is the visit frequency of agent u at location l, W_{total} is the total number of visits, and E_{ul} represents the experienced dissimilarity at location l, calculated as the weighted average of pairwise income dissimilarities between neighborhood u and all other neighborhoods j present at location l. A higher experienced segregation index value indicates a higher degree of social isolation and reduced exposure to diverse socio-economic groups across the city. The full derivation of the pairwise dissimilarity matrix and the computational implementation of E_{ul} are provided in Supplementary Information Section 2.7.

- **Income rank correlation.** To quantify income homogeneity in destination choice, we compute the Pearson correlation coefficient between (i) the income rank of an agent's home CBG and (ii) the average income rank of the neighborhoods the agent visits. The visited-neighborhood income rank is computed as a visit-frequency weighted average so that frequently visited locations contribute proportionally more than occasional trips. Higher positive correlation indicates stronger income-based sorting in daily mobility and lower potential for cross-income contact, whereas values closer to zero indicate weaker stratification and greater mixing across income groups.

Hot Spot Analysis. To generate the hotspot analysis, Point of Interest (POI) visit counts are first spatially aggregated to the Census Block Group level to normalize the data distribution across the study area. We then construct a spatial weight matrix based on Queen contiguity to define neighborhood relationships between adjacent block groups. Using these weights, we perform a Getis-Ord Gi statistic[42] (Hot Spot Analysis) to identify statistically significant spatial clusters of high and low visit density. To ensure statistical rigor, a confidence level of 95% is applied, filtering the results to only include features with a p-value < 0.05 and a positive z-score, thereby isolating clusters of intense activity (hotspots) while excluding random spatial variations.

Data availability

The raw mobility data used in this study were obtained from SafeGraph during the COVID-19 pandemic. Due to licensing restrictions, these raw data cannot be shared publicly, but they are available from the authors upon reasonable request and with approval from SafeGraph. The results of our model can be reproduced without accessing to the Safegraph mobility patterns dataset. All other datasets analyzed in this study are publicly available and are described in the Methods section.

Code availability

All code used to implement the models can be obtained from: GitHub repository (Theory-Informed Generative Agents).

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Acknowledgements

This work was supported in part by start-up funds from the University of Virginia.

Author contributions

H.L. and H.D. conceptualized and designed the study. H.L. collected and processed the data. H.L. and R.L. performed experiments. H.L., Y.L., and H.D. prepared the figures. H.L., Y.L., S.P., and H.D. analyzed the results. H.L. and H.D. wrote the initial draft. A.W., Y.L., S.P., and H.D. provided guidance and feedback for the study. R.L. and H.D. verified the results. All authors prepared the final version of the manuscript.

Competing interests

The authors declare no competing interests.

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